



УНИВЕРЗИТЕТ У БАЊОЈ ЛУЦИ
UNIVERSITY OF BANJA LUKA
ТЕХНОЛОШКИ ФАКУЛТЕТ
FACULTY OF TECHNOLOGY



Nanomaterials Research: A Path from PhD to Academia

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University of Banja Luka, Bosnia and Herzegovina



Seminar at the University of Concepción, Chile
19. January 2026

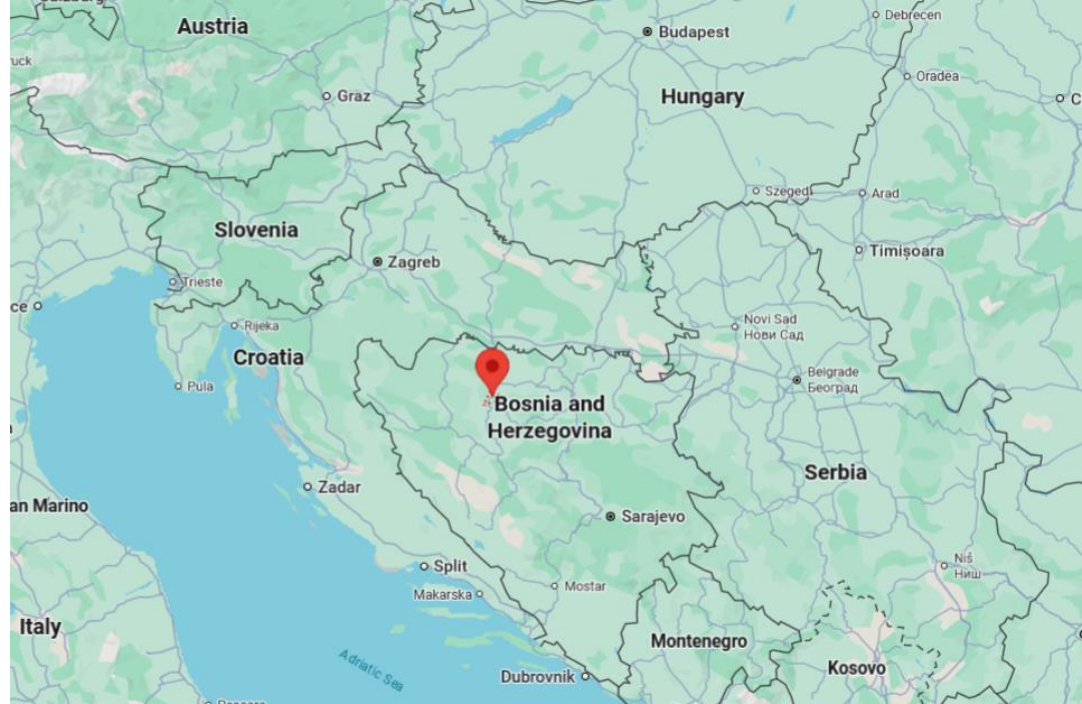
Our distance



Introduction to Banja Luka

Banja Luka is the second-largest city in Bosnia and Herzegovina, known for its cultural heritage and natural beauty.

Thanks to Dr. Z. Gazibarić





Banja Luka – The City of Sports



- Banja Luka hosts notable sports events, including the Belgrade–Banja Luka cycling race , the Banja Luka Challenger tennis tournament , and the Banja Luka Half Marathon.
- In 2024, the city hosted the World Rafting Championships, attracting over 400 competitors from 22 countries.
- Novak Đoković participated in the inaugural Srpska Open held in Banja Luka from April 17 to 23, 2023.



Thanks to Dr. Z. Gazibarić

Banja Luka Fortress (Kastel)

- One of the oldest historical monuments in the city, built during Roman and Ottoman times for defense purposes.



Thanks to
Dr. Z. Gazibarić

Nanomaterials Research

Nanomaterials

-materials that have at least one dimension below 100 nm-

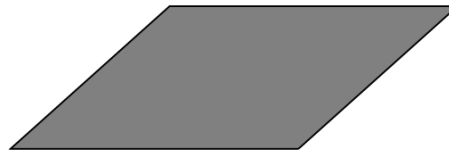
0D



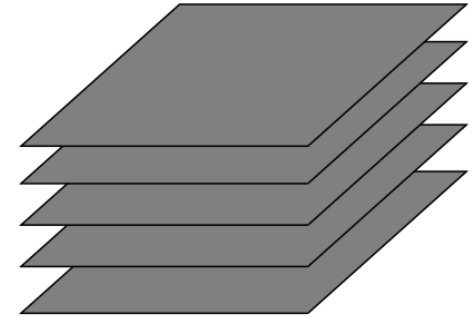
1D



2D



3D



Metal oxides
Fullerenes
Carbon dots
Quantum dots

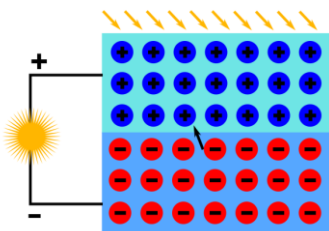
Nanotubes
Nanowires
Nanobelts
Nanofibers

Graphene
Metal sulfides
MXenes
Boron nitride

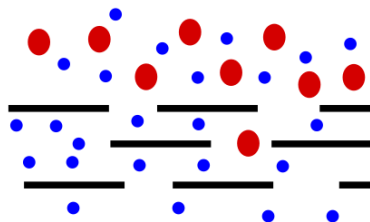
Zeolites
MOF
Active carbon
Composites

Some nanomaterials applications

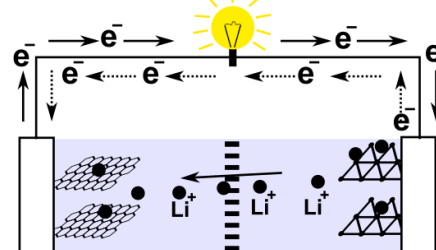
Solar cells



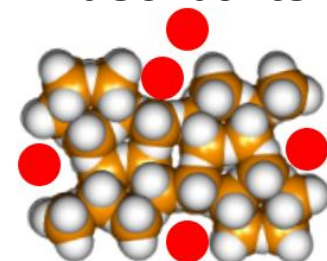
Membranes



Batteries

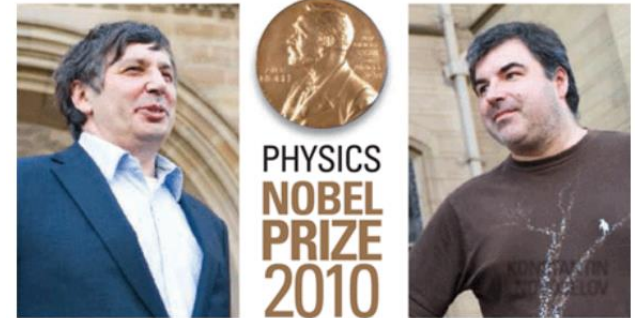
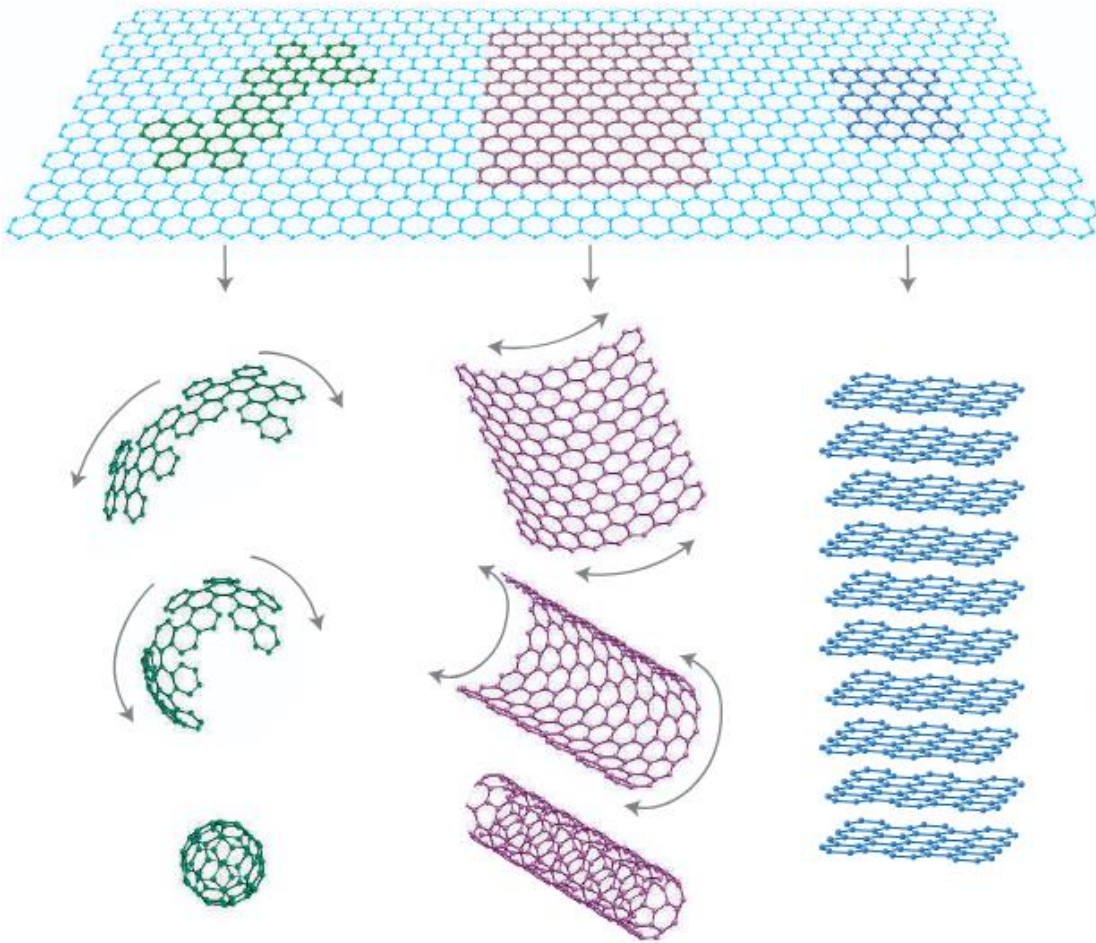


Adsorbents



Graphene nanostructures

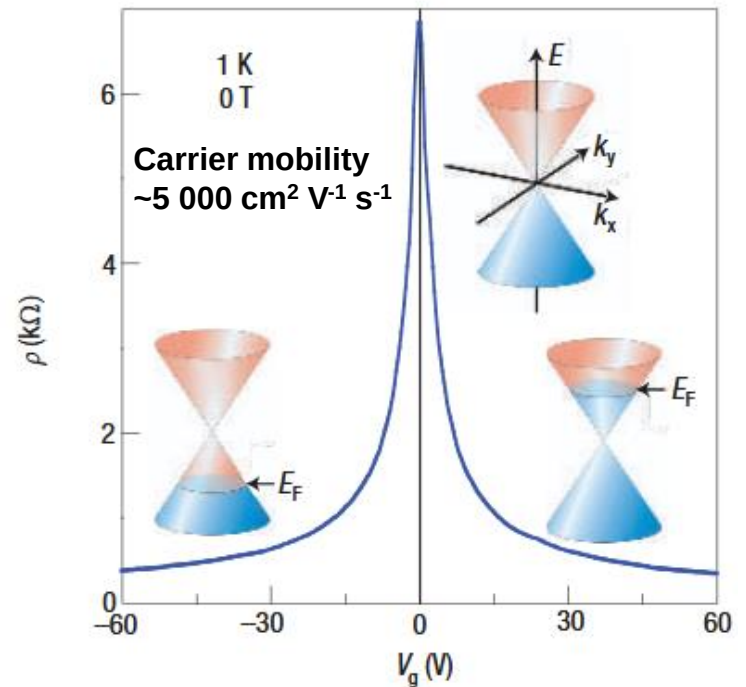
Graphene: mother of all graphitic forms¹



ANDRE GEIM (LEFT); KONSTANTIN NOVOSELOV (RIGHT)

CREDITS: RUSSELL HART/UNIVERSITY OF MANCHESTER; © THE NOBEL FOUNDATION (INSET)

Ambipolar electric field effect in single-layer graphene

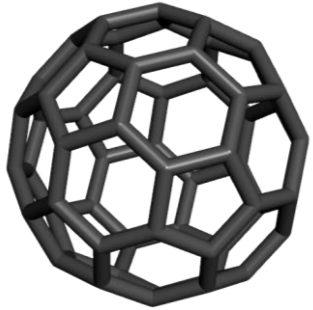


1. Geim, A. K.; Novoselov, K. S. The Rise of Graphene. *Nature Mater* **2007**, 6 (3), 183–191. <https://doi.org/10.1038/nmat1849>.

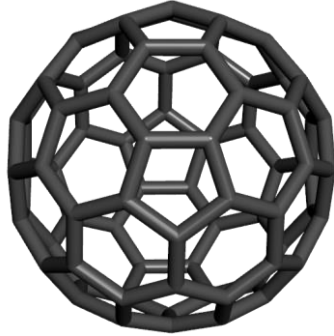
Common nanostructures derived from graphene

Fullerens

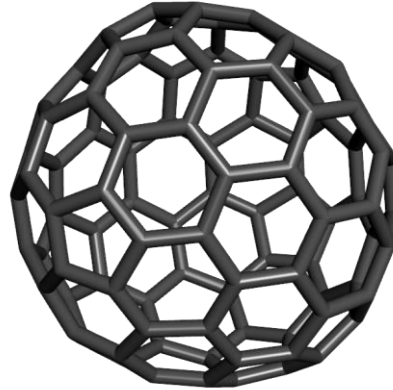
C_{60}



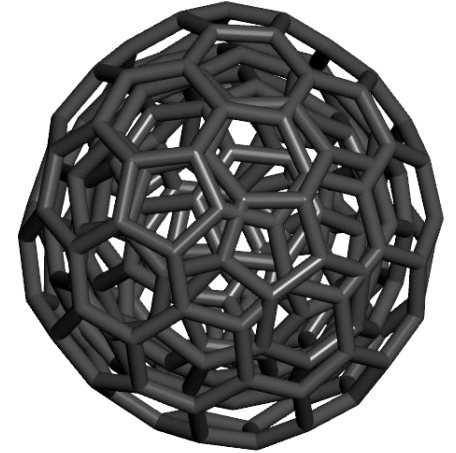
C_{80}



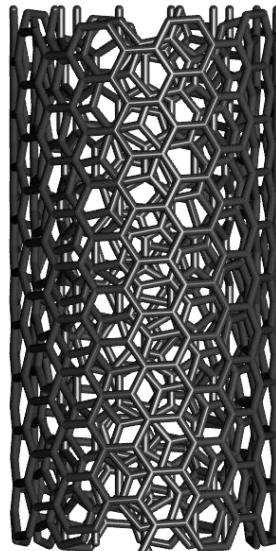
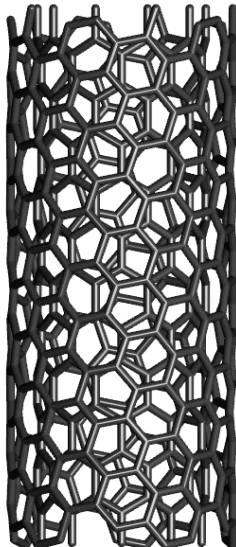
C_{100}



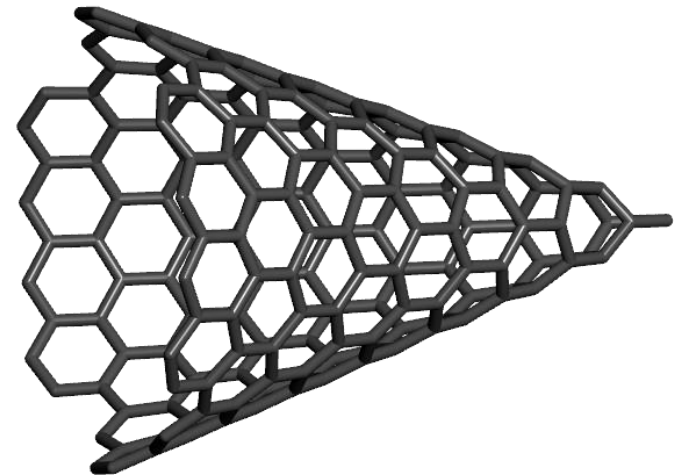
Nanoonion



Carbon nanotubes



Carbon nanohorns



Research conducted in laboratory

PhD students

Master students

Post-doctoral researchers



Source: <https://www.niche.com/blog/4-methods-to-get-into-a-research-lab/>

Research interests and activities

Zeolites

Porosity and structural stability

Carbon nanotubes and dispersants

Sol-gel dispersants for SWCNT

Electrically conducting thin films

Graphene

Gas separation membranes

Electrically conducting graphene

Scaffolds for bone recovery

Drug adsorption and delivery

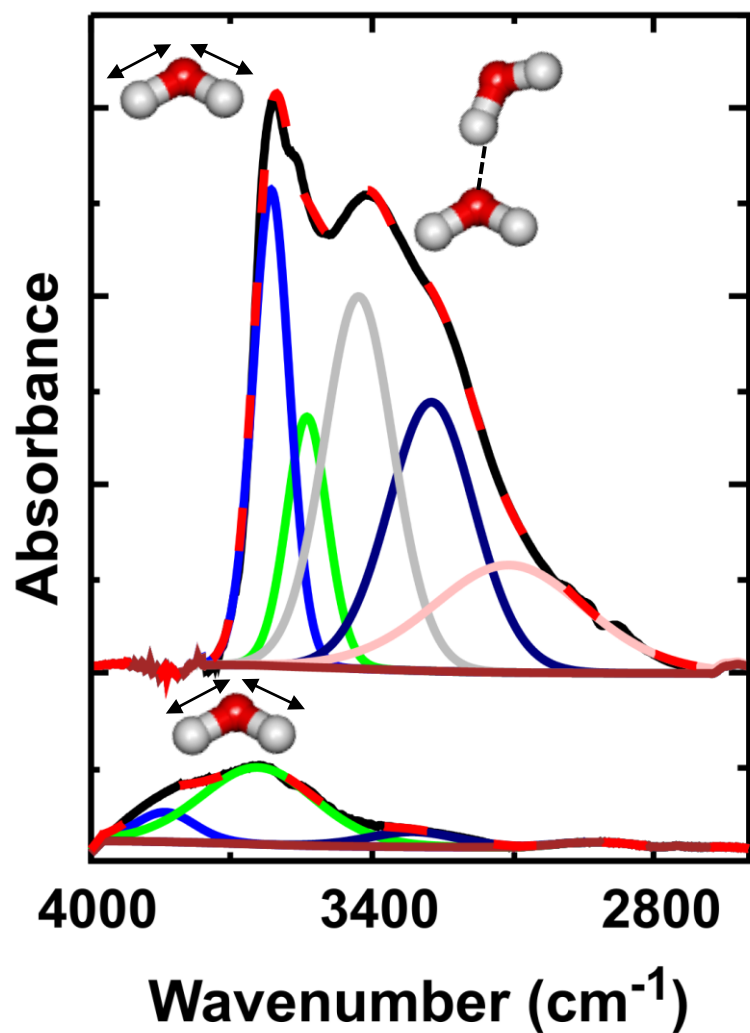
Adsorption of letrozole drug onto clinoptilolite surface

Zeolite/apatite biocompatible coatings

Coating the Ti-alloy implants for anticorrosion

Water desorption from CLI zeolites

Infrared spectra

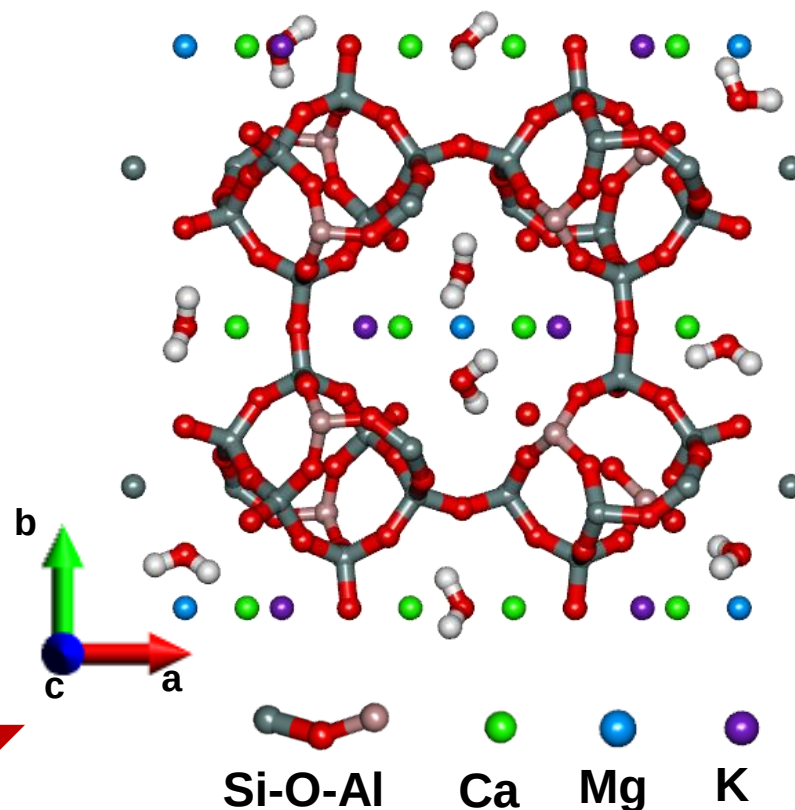


378 K

H₂O desorption

1073 K

Clinoptilolite

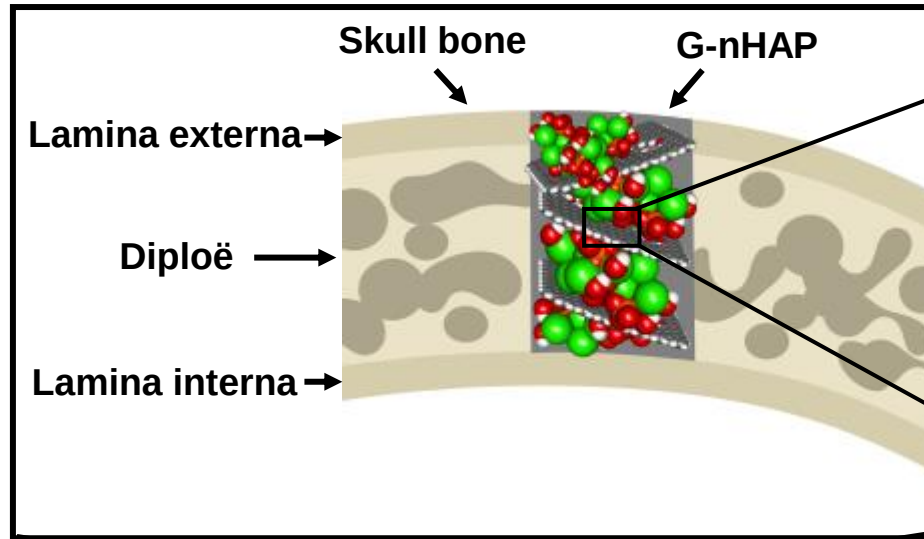


$$S_{\text{BET}} = 60 \text{ m}^2 \text{ g}^{-1}$$

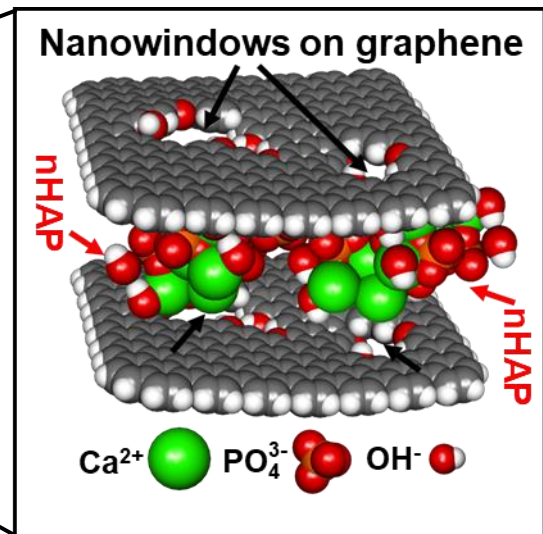
$$S_{\alpha\text{S}} = 80 \text{ m}^2 \text{ g}^{-1}$$

G-nHAP scaffold for successful bone healing

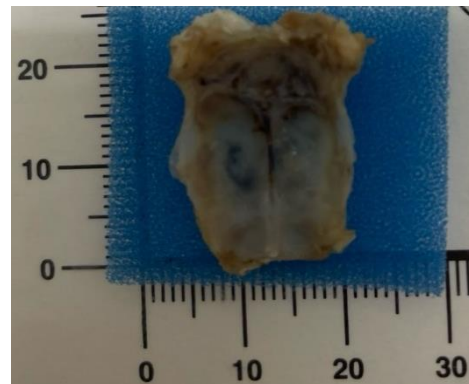
Bone filled with G-nHAP



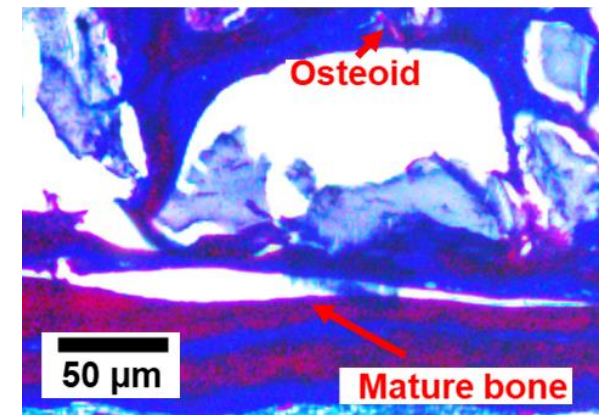
G-nHAP



Calvaria after 6 weeks



Optical micrograph Masson trichrome stained

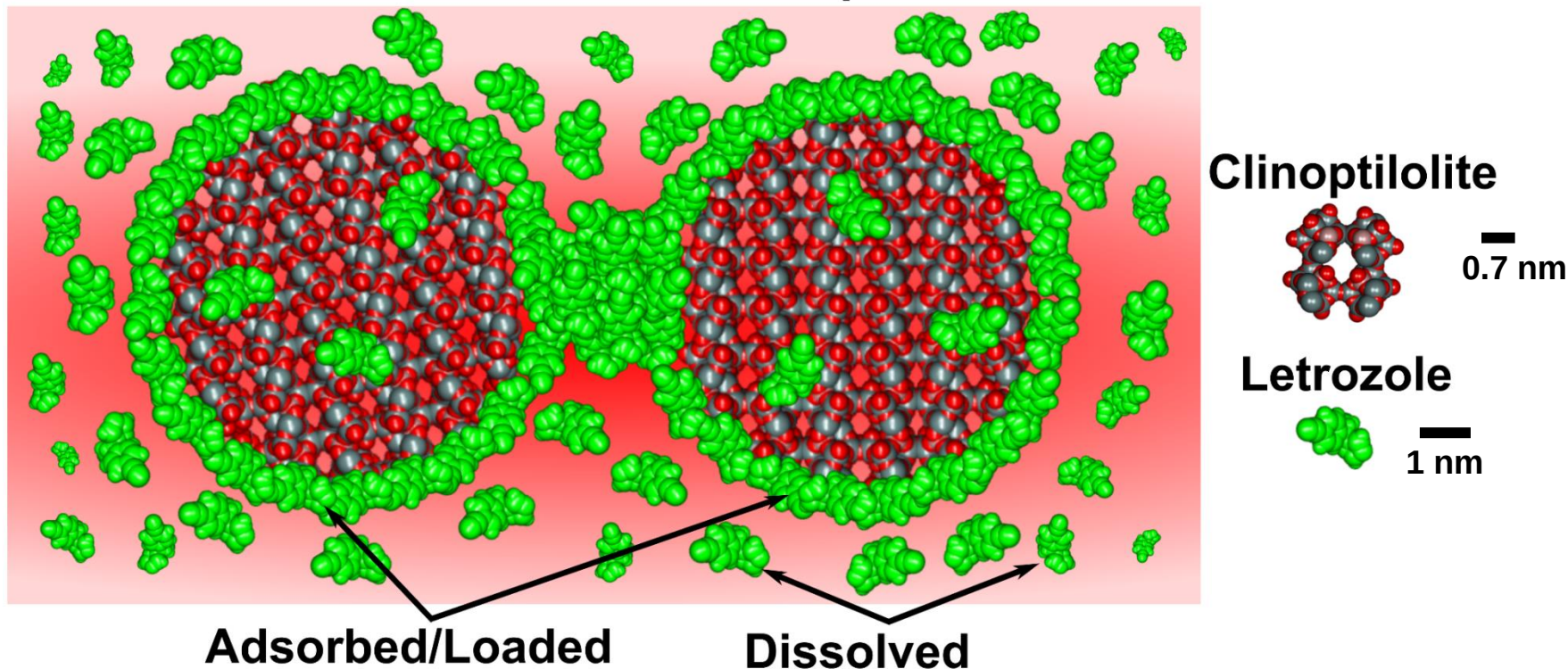


The G-nHAP helps a fast bone tissue regeneration

Kukobat, R.; et al., *Carbon* **2025**, 243, 120589.
<https://doi.org/10.1016/j.carbon.2025.120589>

Our main approach to drug dissolution

Letrozole dissolution from clinoptilolite



Importance of drug dissolution

Poorly soluble drugs may not adsorb to the organism, leading to the treatment failure

Approach in this study

Experiments: Adsorption/release
Theory: Adsorption energy using DFT method

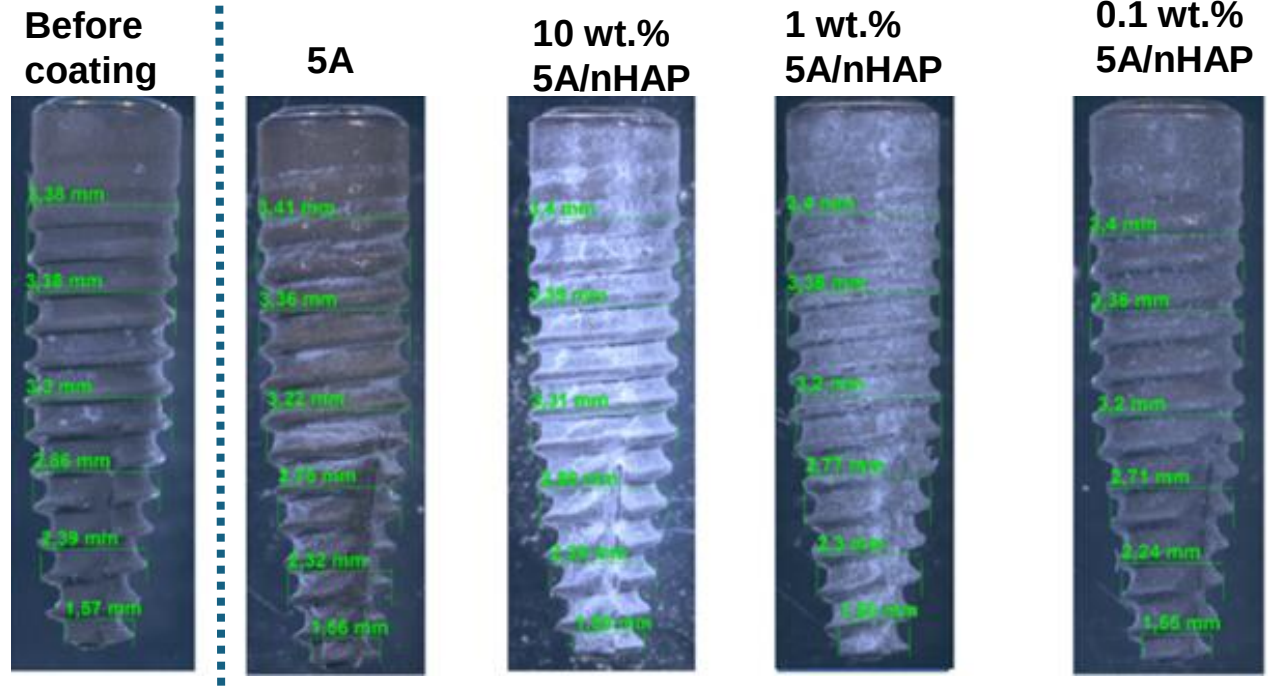
Zeolite-nHAP coatings on implants

Corroded implants



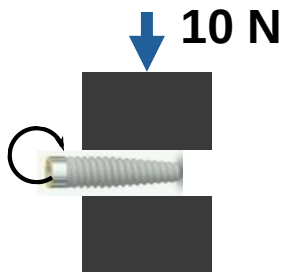
<https://drsehmi.co.uk/metal-showing-on-a-dental-implant-one-possible-solution/>

Ti/Zr implants coated with zeolite/nHAP



Adhesion test

Implant between the tissues



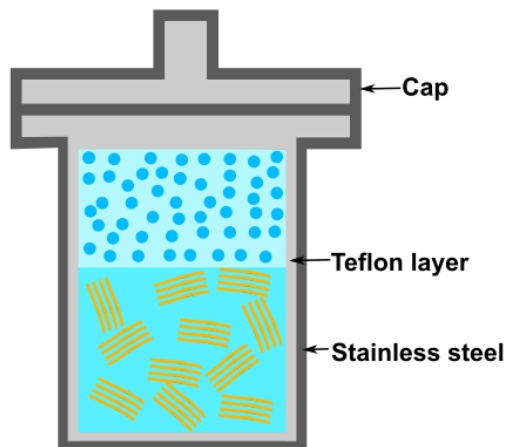
Thanks to
Ms. Mila
Vukoman,
master student

The percent of the coating removal

Material	Removal percent (wt.%)
5A	100.00
0.1 wt.% 5A-nHAP	43.75
1.0 wt.% 5A-nHAP	38.12
10.0 wt.% 5A-nHAP	60.25

Hydrothermal reduction of GO

Autoclave for GO reduction

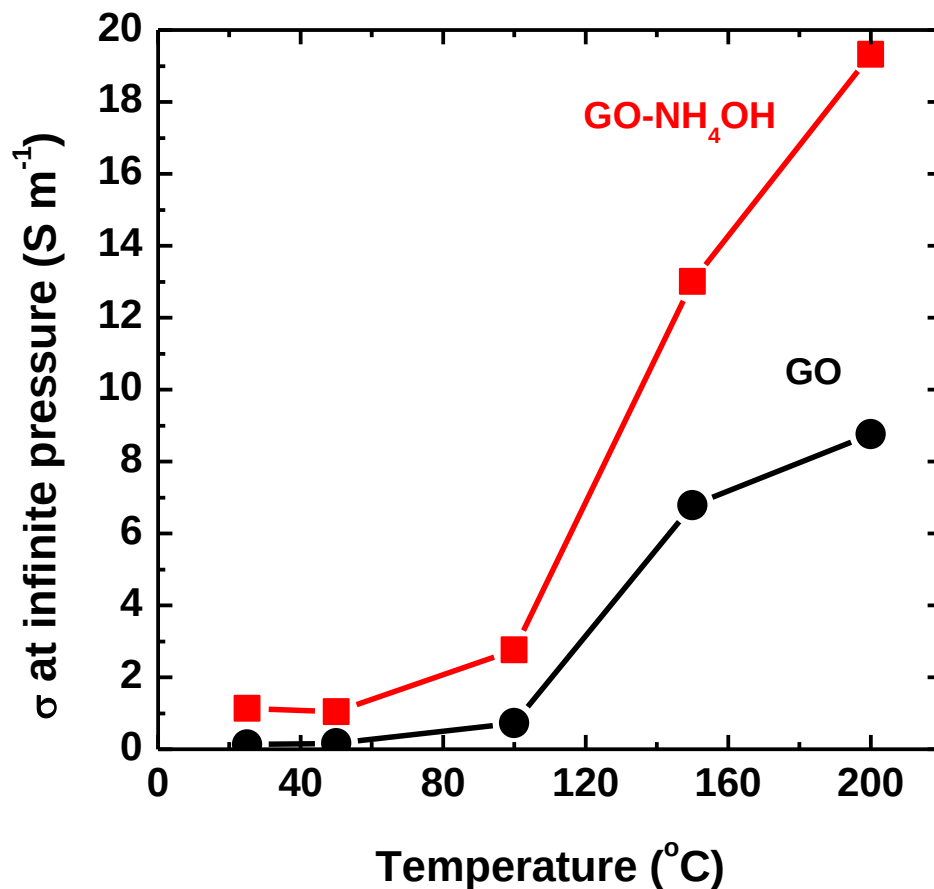


GO colloidal dispersions

25 °C 50 °C 100 °C 150 °C 200 °C

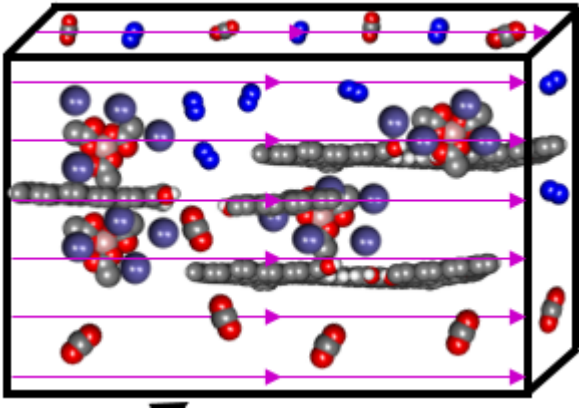


Electrical conductivity vs. temperature



Graphene-based membranes for CO₂ separation

Aligned graphene-membrane

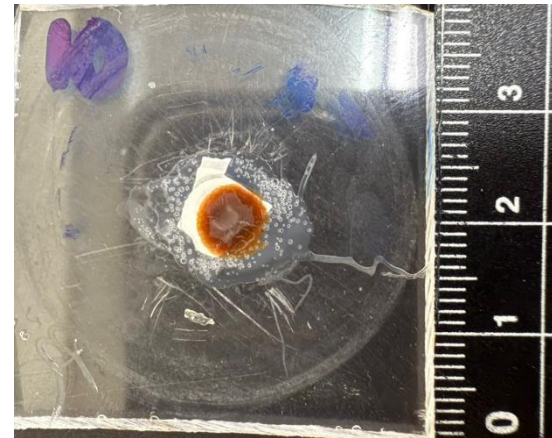


Permeance of CO₂

t, min	P, mol/m ² sPa
8.13	7.97E-10
22.08	2.93E-10
9.5	6.81875E-10
11.56	5.60E-10

Average CO ₂	5.83E-10
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Membrane top view



Permeance of N₂

t, min	P, mol/m ² sPa
187.3	6.65E-13

$$Selectivity = \frac{CO_2}{N_2} = 880$$

Thanks to Ms. Sarai Marquez Garcia,
master student

Who does research and where research is conducted

Students

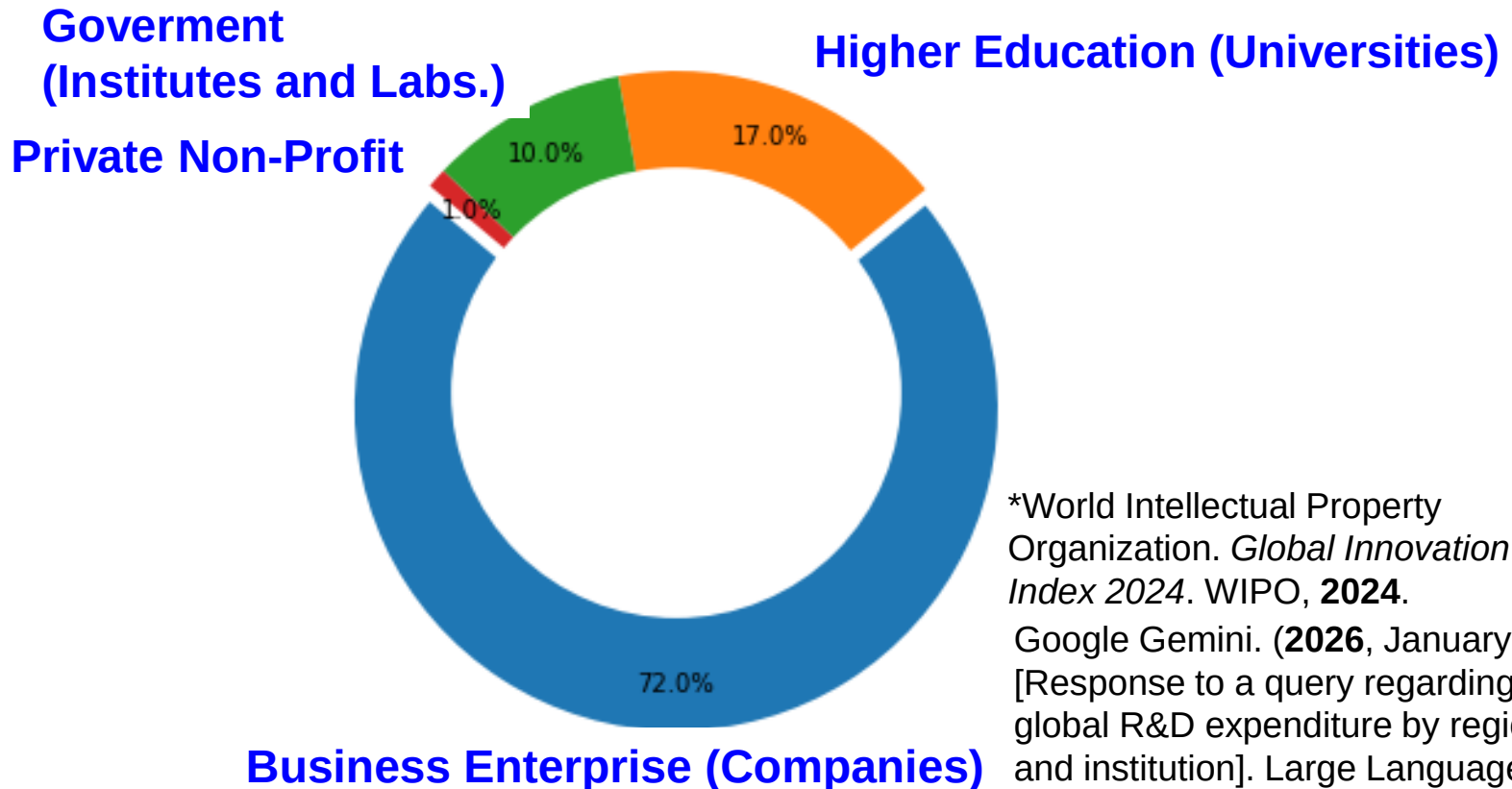
**Bachelor
Master
Doctoral**

Researchers

**Professionals
Post-docs**

Academic stuff

Professors



*World Intellectual Property Organization. *Global Innovation Index 2024*. WIPO, **2024**.

Google Gemini. (**2026**, January 13). [Response to a query regarding global R&D expenditure by region and institution]. Large Language Model.

Responsibilities of PhD and master students

**Meeting
supervisor**

**Reading
literature**

**Attending
group meetings**



**Writing up
results**

**Reporting and
communicating**

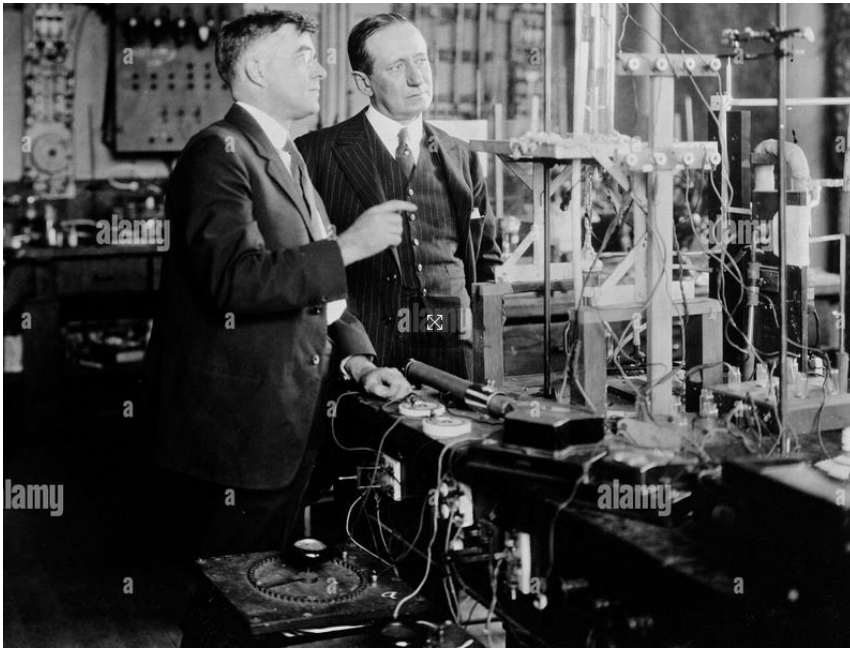
**Analyzing literature
and results**

Being a Researcher

Research is one of the most challenging professions

Langmuir and Markoni (1922)

General Electric Research Laboratory in New York



Izvor: <https://www.alamy.com/stock-photo-langmuir-and-marconi-in-the-general-electric-research-laboratory-new-104002884.html>

Characteristics that scientists should have

- Honesty**
- Diligence**
- Strong will**
- Patience**
- Intense attention**
- Deep concentration**
- Knowing how to think**
- Discipline and accuracy**
- Creative imagination**
- Solid general culture**
- Solid professional education**
- Mastery of a given field**
- Knowledge of research methodology**

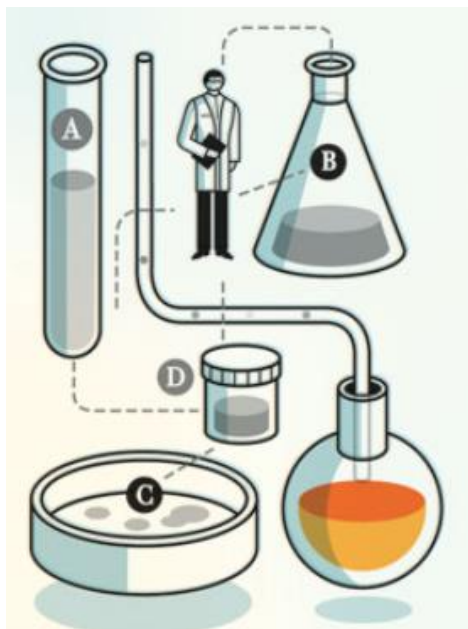
Being researcher is challenging

What is a scientific research

Scientific research is an objective and planned examination of a problem according to certain methodological rules.¹

1. *Frascati Manual. The Measurement of Scientific, Technological and Innovation Activities.* doi:[10.1787/9789264239012-en](https://doi.org/10.1787/9789264239012-en).

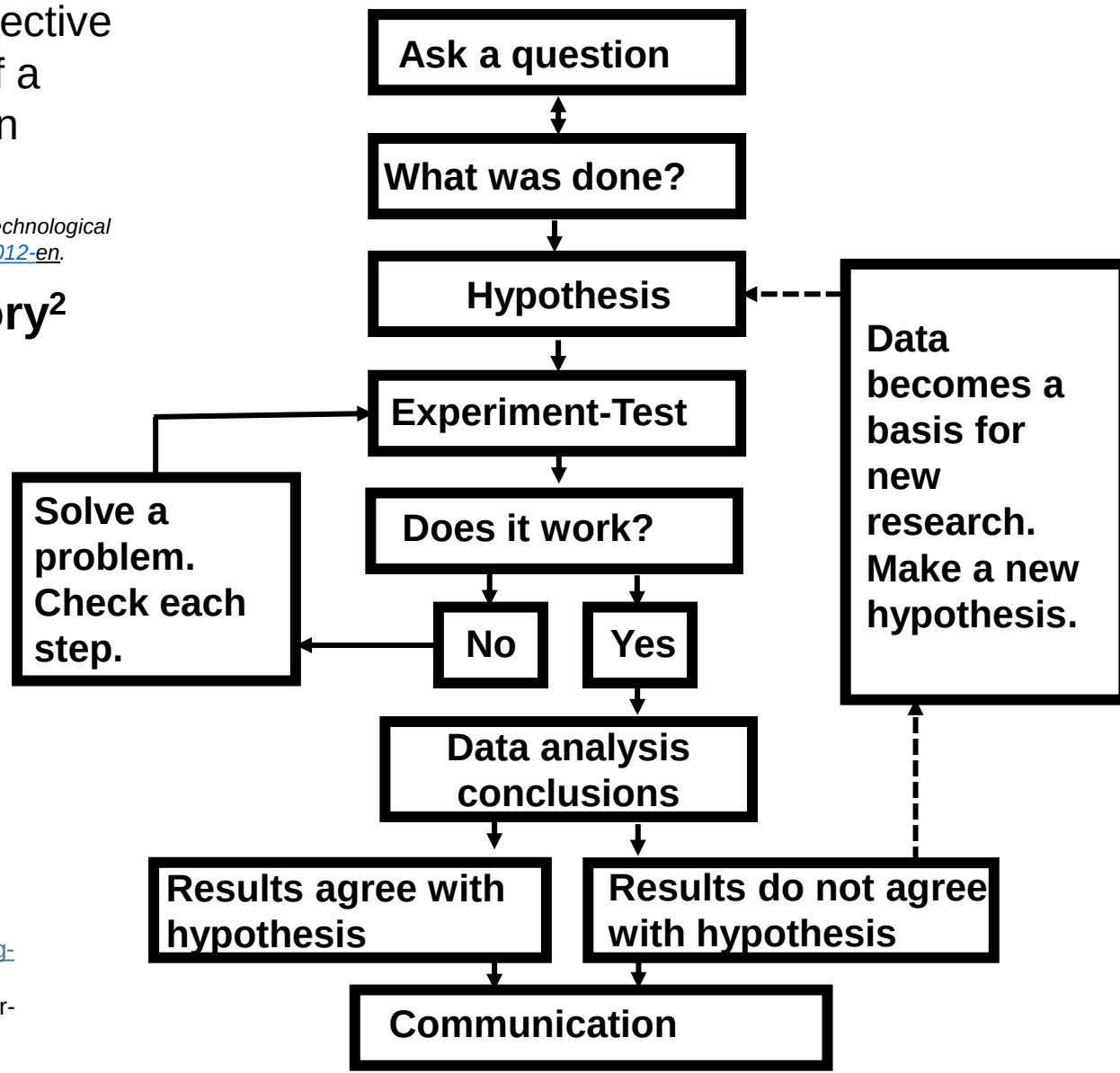
Researcher in a laboratory²



2. Izvor: <https://www.sciencenews.org/article/redoing-scientific-research-best-way-find-truth>

3. Izvor: <https://www.sciencebuddies.org/science-fair-projects/science-fair/steps-of-the-scientific-method>

Phases of scientific research³



Writing and publishing papers



****<https://lp2m.uma.ac.id/2021/01/05/tips-for-knowing-a-good-and-correct-scientific-journal-format/>**

Responsibilities of Professors

**Managing
workflow**

**Training new
hires**

Teaching



**Crating and
managing
team schedules**

**Writing
project proposals**

**Writing project
reports**

Challenges: Funding

Horizon Europe is the ambitious EU research & innovation framework programme for 2021-2027 with a budget of €95.5 billion.



Pillar 1 Excellent Science

European Research Council

Marie Skłodowska-Curie
Actions

Research Infrastructures

Pillar 2 Global Challenges and European Industrial Competitiveness

- Clusters
- Health
 - Culture, Creativity and Inclusive Society
 - Civil Security for Society
 - Digital, Industry and Space
 - Climate, Energy and Mobility
 - Food, Bioeconomy, Natural Resources, Agriculture and Environment

Joint Research Centre

Pillar 3 Innovative Europe

European Innovation Council

European innovation
ecosystems

European Institute of
Innovation
and Technology

Widening Participation and Strengthening the European Research Area

Widening participation and spreading excellence

Reforming and Enhancing the European R&I system

Elements for the project preparation



A Letter of Interest

A cover letter of sorts to present yourself/company, track record and why you are suited for the task



Background and introduction

Your understanding of the project and the scope of work



Evaluation plan/design matrix

Includes the evaluation questions, methodology/approach, data collection methods and means of analysis



Team/organisational capacity

Professional profile of the individual/team composition and the added value each person brings



Timelines

Tentative schedule for the implementation of activities and deliverables



Budget

Professional fees and associated costs

Necessary skills



Budget your calendar

Prioritize your time and don't overschedule yourself.



Delegate effectively

Hand off tasks to free up your time for high-priority items.



Communicate and collaborate

Work with your team to stay organized and successful.



Leverage software features

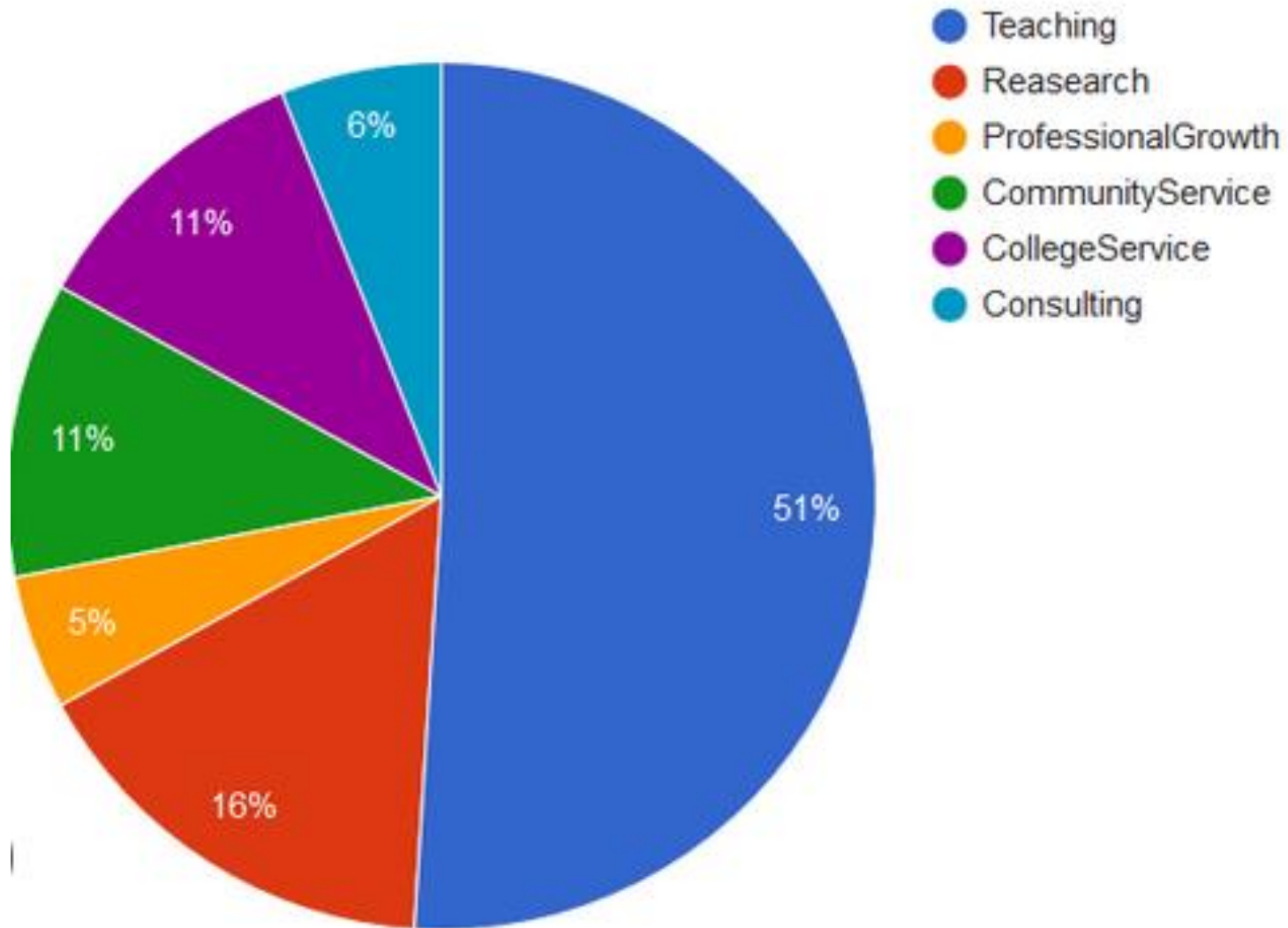
Get to know software features so you can save time on manual work.



Be a team player

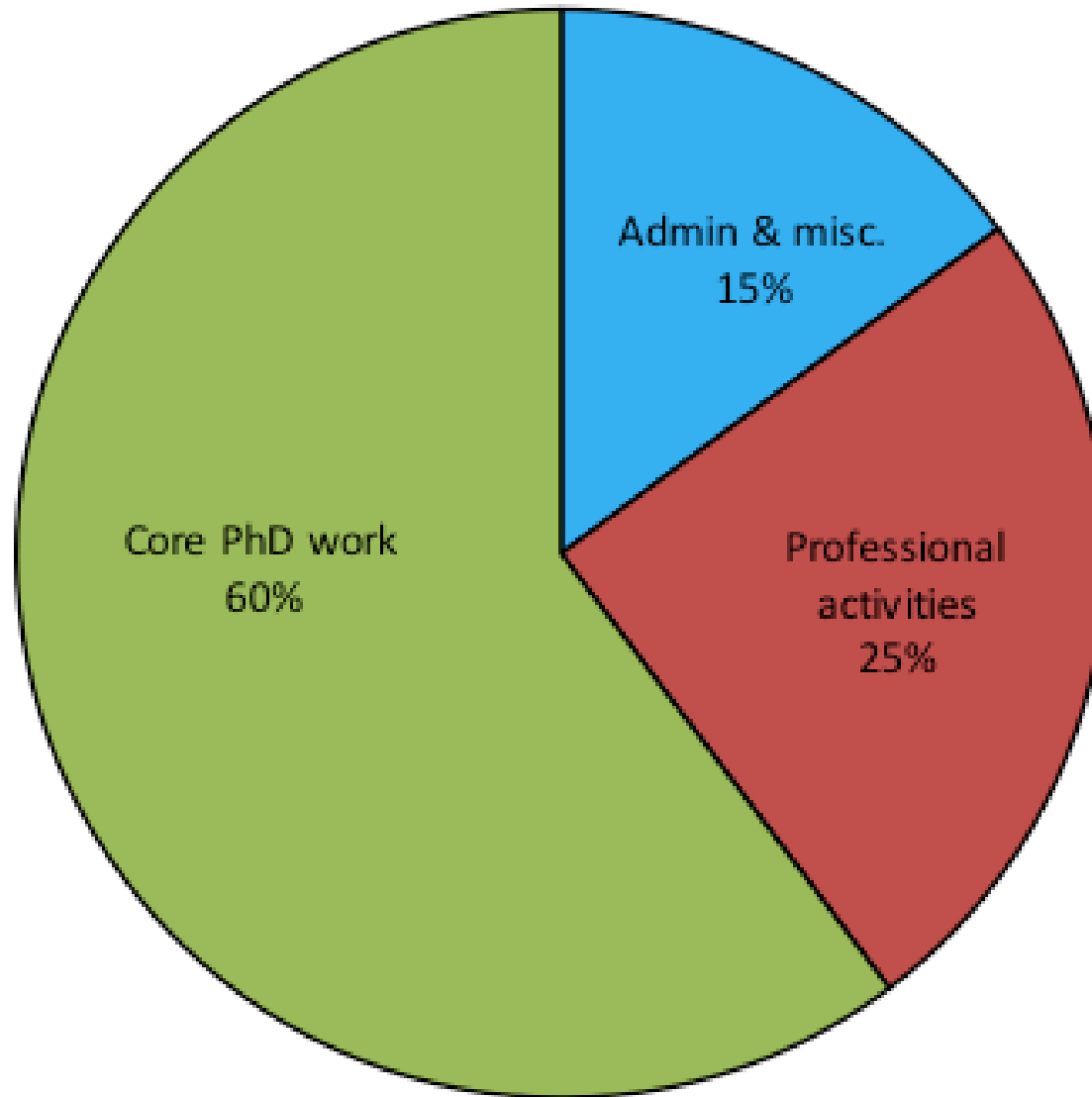
Achieve your personal and team goals faster by working as a team.

Distribution of time for university professor



Brase, C. H., & Brase, C. P. (2018). *Understandable statistics: Concepts and methods* (12th ed.). Cengage Learning

Distribution of time for a PhD student



Patterson, J. (2016, May 11). How long does it take to do a PhD. *The Thesis Whisperer*.
<https://thesiswhisperer.com/2016/05/11/how-long-does-it-take-to-do-a-phd/>

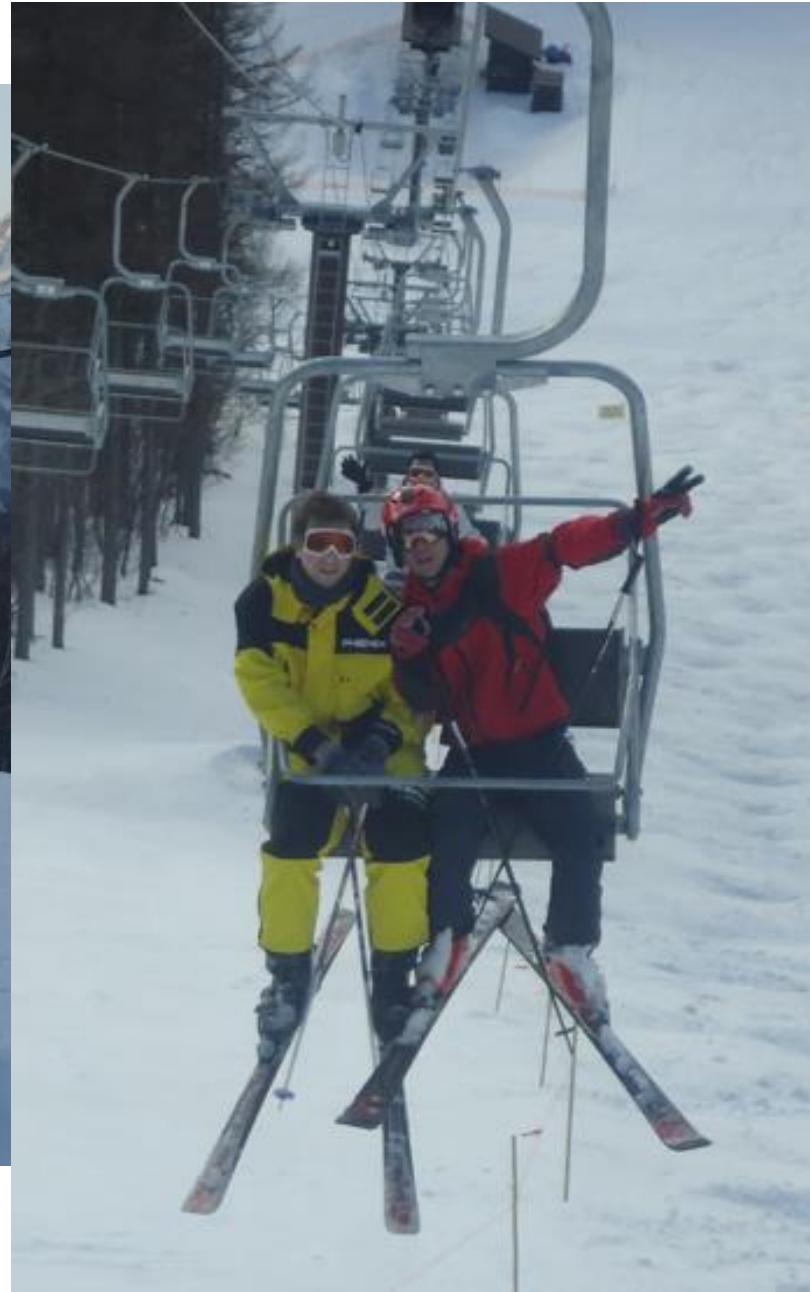
Summary

Working as a PhD student is guided by a supervisor and requires time to accomplish hypothesis

Post doctoral research and professional researchers is more independent

Working as a professor requires more administrative work, writing projects and does not leave much time for scientific work

Memories with Nando



Thank you for your attention